Solving Trigonometric Equations (TE3)

Math 172 lecture notes · Section 4.4.3

Learning objective: find all solutions to a trigonometric equation.

Try each part before you open the answer. Your thinking is what makes it stick.

Problem from your notes Text from the handout Extra note (not in your handout)

Overview

FROM THE HANDOUT

Now that we've learned about properties of and identities involving trigonometric functions, we will now turn to solving equations that include trigonometric functions.

Activity 1

Activity 1 · from your notes

Consider the equation $2 \sin \theta - 1 = 0$.

(a) *What first step would you take to isolate $\sin \theta$?*

Hint: *If given the equation $2x - 1 = 0$, what would your first step be?*

Answer (a)

Add 1 to both sides: $2 \sin \theta = 1$. It's the same first move as for $2x - 1 = 0$.

(b) *Continue using algebra to isolate $\sin \theta$. What equation would you have left?*

Answer (b)

Divide both sides by 2.

$$\sin \theta = \frac{1}{2}$$

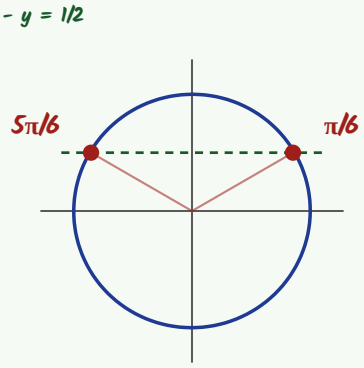
(c) For what values of θ, on the interval $[0, 2\pi)$, is the equation true?

> **Hint:** Use the unit circle to help you find all the values of θ.

Answer (c)

Sine is the y-coordinate on the unit circle. Draw the line $y = \frac{1}{2}$: it hits the circle twice, in Quadrants I and II.

$$\theta = \frac{\pi}{6}, \quad \frac{5\pi}{6}$$



Note

FROM THE HANDOUT

One way to solve a trigonometric equation is to **isolate the trigonometric term** on one side. Once it has been isolated, you can then determine values of θ that satisfy the equation.

Definition

FROM THE HANDOUT

General solution. In many cases, you will be asked to find the solution to a trigonometric equation within a given interval (such as $[0, 2\pi)$). If, however, you are asked to find the general solution, you need to include all solutions: every angle coterminal with the solutions in $[0, 2\pi)$.

If α is a solution, the general solution is $\theta = \alpha + 2\pi n$, where n is any integer.

(The handout writes $n \in \mathbb{Z}$, where $\mathbb{Z}$ is the set of integers.)

Activity 2 · from your notes

From Activity 1, $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ satisfy $2\sin\theta - 1 = 0$ on $[0, 2\pi)$. What other values of θ also work?

(a) If $\frac{\pi}{6}$ is a solution, is $\frac{\pi}{6} + 2\pi$ a solution?

> **Hint:** One complete revolution is 2π, so the positive x-axis can be 0 or 2π radians (or 4π, or -2π, ...).

Answer (a)

Adding 2π is one full trip around the circle, so you land on the same point. $\sin(\frac{13\pi}{6}) = \frac{1}{2}$, and $2(\frac{1}{2}) - 1 = 0$.

Yes

(b) Is $\frac{\pi}{6} - 2\pi$ a solution?

Answer (b)

Subtracting 2π is one full turn clockwise, which lands on the same point: $\sin(\frac{-11\pi}{6}) = \frac{1}{2}$.

Yes

(c) Is $\frac{\pi}{6} + \frac{\pi}{2}$ a solution?

Answer (c)

$\frac{\pi}{6} + \frac{\pi}{2} = \frac{2\pi}{3}$. That's a quarter turn, not a full one, so we land on a different point: $\sin(\frac{2\pi}{3}) = \frac{\sqrt{3}}{2}$, not $\frac{1}{2}$.

No

Your handout prints $\frac{\pi}{6} + 2\pi$ here, which repeats part (a). The textbook version asks about $\frac{\pi}{6} + \frac{\pi}{2}$, so that's the one we use.

(d) What other solutions can you find to $2 \sin\theta - 1 = 0$?

Answer (d)

Any angle that lands on one of the two points, for example $\frac{13\pi}{6}$, $\frac{17\pi}{6}$, $\frac{25\pi}{6}$, $\frac{-7\pi}{6}$, $\frac{-11\pi}{6}$, …

(e) Use parts (a) and (b) to find the general solution to $2 \sin\theta - 1 = 0$.

Hint: Any angle coterminal with either solution also works, so add or subtract any multiple of 2π.

Answer (e)

$$\theta = \frac{\pi}{6} + 2\pi n \quad \text{and} \quad \theta = \frac{5\pi}{6} + 2\pi n, \quad n \text{ any integer}$$

Activity 3

Activity 3 · from your notes

Consider the equation $(2 \sin\theta + \sqrt{3})(\cos\theta - 1) = 0$.

(a) Refer back to the Zero Product Property. What equations do we need to solve?

Answer (a)

If a product is zero, at least one factor is zero:

$$2 \sin\theta + \sqrt{3} = 0 \quad \text{or} \quad \cos\theta - 1 = 0$$

(b) Solve $2 \sin\theta + \sqrt{3} = 0$ on $[0, 2\pi)$.

Answer (b)

$\sin\theta = -\frac{\sqrt{3}}{2}$. Negative y, so Quadrants III and IV, with reference angle $\frac{\pi}{3}$.

$$\theta = \frac{4\pi}{3}, \quad \frac{5\pi}{3}$$

(c) Solve $\cos\theta - 1 = 0$ on $[0, 2\pi)$.

Answer (c)

$\cos\theta = 1$ where $x = 1$, at the far right of the circle.

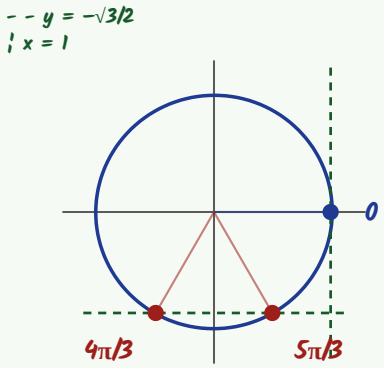
$$\theta = 0$$

> ⚠ **Watch out:** 2π lands on the same point, but the interval $[0, 2\pi)$ leaves out 2π.

(d) Combine parts (b) and (c). What are all the solutions on $[0, 2\pi)$?

Answer (d)

$$\theta = 0, \ \frac{4\pi}{3}, \ \frac{5\pi}{3}$$



(e) Find the general solution.

Answer (e)

$$\theta = 2\pi n, \ \theta = \frac{4\pi}{3} + 2\pi n, \ \theta = \frac{5\pi}{3} + 2\pi n, \ n \text{ any integer}$$

Activity 4

Activity 4 · from your notes

Trig equations can include other trig functions. Consider $4 = 5 + \tan\theta$.

(a) Use algebra to isolate $\tan\theta$.

Answer (a)

Subtract 5 from both sides.

$$\tan\theta = -1$$

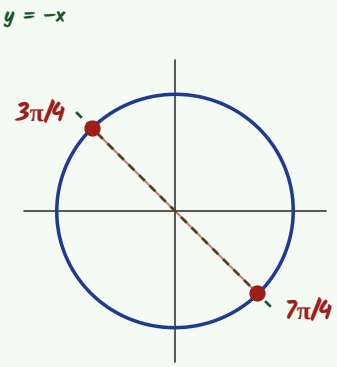
(b) For what values of θ on $[0, 2\pi)$ is the equation true?

Hint: Recall that $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

Answer (b)

$\tan\theta = -1$ means $\sin\theta$ and $\cos\theta$ are the same size with opposite signs. That happens in Quadrants II and IV, with reference angle $\dfrac{\pi}{4}$.

$$\theta = \frac{3\pi}{4},\ \frac{7\pi}{4}$$



(c) What is the general solution of $4 = 5 + \tan\theta$?

Answer (c)

$$\theta = \frac{3\pi}{4} + 2\pi n \quad \text{and} \quad \theta = \frac{7\pi}{4} + 2\pi n$$

The two solutions are exactly π apart, so this can also be written as one family: $\theta = \dfrac{3\pi}{4} + \pi n$, n any integer.

Activity 5

Activity 5 · from your notes

Consider the equation $(\sin\theta - 1)(\csc\theta + 2) = 0$.

(a) Refer back to the Zero Product Property. What two equations do we solve?

Answer (a)

$\sin\theta - 1 = 0 \quad \text{or} \quad \csc\theta + 2 = 0$

(b) Solve $\sin\theta - 1 = 0$ on $[0, 2\pi)$.

Answer (b)

$\sin\theta = 1$ at the top of the circle.

$$\theta = \frac{\pi}{2}$$

(c) To solve $\csc\theta + 2 = 0$, isolate $\csc\theta$. What does $\csc\theta$ equal?

Answer (c)

$$\csc\theta = -2$$

(d) Recall that $\csc\theta = \frac{1}{\sin\theta}$. If $\csc\theta = -2$, what does $\sin\theta$ equal?

Answer (d)

Flip both sides (take reciprocals).

$$\sin\theta = -\frac{1}{2}$$

(e) What values of θ satisfy $\csc\theta = -2$ on $[0, 2\pi)$?

Answer (e)

Negative y: Quadrants III and IV, with reference angle $\frac{\pi}{6}$.

$$\theta = \frac{7\pi}{6}, \quad \frac{11\pi}{6}$$

(f) Combine parts (b) and (e). What are all the solutions on $[0, 2\pi)$?

Answer (f)

$$\theta = \frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$$

(g) Find the general solution.

Answer (g)

$$\theta = \frac{\pi}{2} + 2\pi n,\ \frac{7\pi}{6} + 2\pi n,\ \frac{11\pi}{6} + 2\pi n,\ n \text{ any integer}$$

Activity 6

Activity 6 · from your notes

Solve each equation on the interval $[0, 2\pi)$.

(a) $3 + \sin\theta = 2$

Answer (a)

Subtract 3: $\sin\theta = -1$, the bottom of the circle.

$$\theta = \frac{3\pi}{2}$$

(b) $\sin\theta(\tan\theta - 1) = 0$

Hint: Recall that $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

Answer (b)

Zero Product Property:

- $\sin\theta = 0$ gives $\theta = 0, \pi$
- $\tan\theta = 1$ gives $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ (sin and cos equal: Quadrants I and III)

$$\theta = 0,\ \frac{\pi}{4},\ \pi,\ \frac{5\pi}{4}$$

Note

FROM THE HANDOUT

In some cases, it is not easy to isolate the trigonometric term like we have been doing so far. The next few activities will help us learn how to apply methods we have used before when solving algebraic equations to solve trigonometric equations.

Activity 7

Activity 7 · from your notes

Consider the equation $4\sin^2\theta - 2 = 0$.

(a) Use algebra to isolate the $\sin^2\theta$ term.

Answer (a)

Add 2, then divide by 4.

$$\sin^2\theta = \frac{1}{2}$$

(b) Take the square root of both sides. What does $\sin\theta$ equal?

Answer (b)

$$\sin\theta = \pm\sqrt{\left(\frac{1}{2}\right)} = \pm\frac{1}{\sqrt{2}}$$

⚠ **Watch out:** Square roots come with $\pm$. Dropping the minus sign loses half the answers.

(c) Which form of the values from part (b) do you see on the unit circle?

Answer (c)

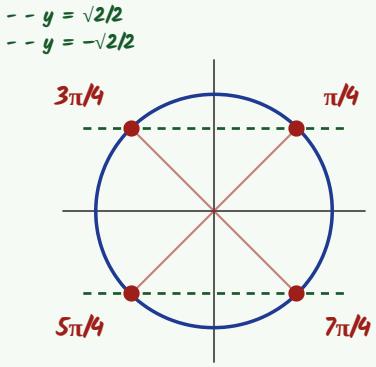
Rationalize: $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. The unit circle uses the form $\pm\frac{\sqrt{2}}{2}$.

(d) *Two values of $\sin\theta$ give four values of θ. What are they on $[0, 2\pi)$?*

Answer (d)

$\sin\theta = \frac{\sqrt{2}}{2}$ gives $\frac{\pi}{4}, \frac{3\pi}{4}$. $\sin\theta = -\frac{\sqrt{2}}{2}$ gives $\frac{5\pi}{4}, \frac{7\pi}{4}$.

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$



Activity 8

Activity 8 · from your notes

Consider the equation $3\sec^2\theta - 4 = 0$.

(a) *Using algebra, isolate the $\sec^2\theta$ term.*

Answer (a)

$$\sec^2\theta = \frac{4}{3}$$

(b) *Use the square root property to solve for $\sec\theta$.*

Answer (b)

$$\sec\theta = \pm\frac{2}{\sqrt{3}} = \pm\frac{2\sqrt{3}}{3}$$

(c) Recall $\sec\theta = \frac{1}{\cos\theta}$. What does $\cos\theta$ equal?

Answer (c)

Take reciprocals.

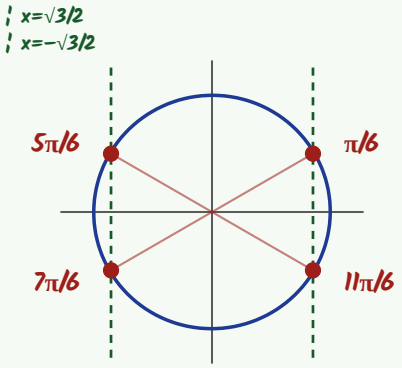
$$\cos\theta = \pm\frac{\sqrt{3}}{2}$$

(d) What values of θ satisfy the equation on $[0, 2\pi)$?

Answer (d)

$\cos\theta$ is the x-coordinate. $\cos\theta = \frac{\sqrt{3}}{2}$: $\frac{\pi}{6}$, $\frac{11\pi}{6}$. $\cos\theta = -\frac{\sqrt{3}}{2}$: $\frac{5\pi}{6}$, $\frac{7\pi}{6}$.

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$



Note

FROM THE HANDOUT

Notice in previous examples, we had to use the square root property to isolate the trigonometric term. The next few activities will highlight another tool to solve trigonometric equations.

Activity 9

Activity 9 · from your notes

Solve by factoring: $2\sin\theta\cos\theta - \sqrt{2}\cos\theta = 0$.

(a) Start with the GCF (Greatest Common Factor). What is the GCF of the terms?

Answer (a)

$\cos\theta$

(b) *Factor out the GCF. What equation do you now have?*

Answer (b)

$\cos\theta\,(2\sin\theta - \sqrt{2}) = 0$

(c) *Use the Zero Product Property. What two equations do you now have?*

Answer (c)

$\cos\theta = 0 \quad \text{or} \quad 2\sin\theta - \sqrt{2} = 0$

(d) *Solve both equations. What values of θ work on $[0, 2\pi)$?*

Answer (d)

$\cos\theta = 0$ gives $\frac{\pi}{2}, \frac{3\pi}{2}$. $\sin\theta = \frac{\sqrt{2}}{2}$ gives $\frac{\pi}{4}, \frac{3\pi}{4}$.

$$\theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}$$

⚠ **Watch out:** *Don't divide both sides by $\cos\theta$. That throws away the solutions where $\cos\theta = 0$.*

Activity 10

Activity 10 · from your notes

Consider the equation $2\cos^2\theta - 5\cos\theta - 3 = 0$.

(a) *It looks like a quadratic! Replace $\cos\theta$ with x: $2x^2 - 5x - 3 = 0$. Factor the left side.*

Answer (a)

$(2x + 1)(x - 3)$

(b) Use the Zero Product Property to write two equations in terms of cos θ.

Answer (b)

$$2\cos\theta + 1 = 0 \quad \text{or} \quad \cos\theta - 3 = 0$$

(c) Solve each equation. What values of θ work on $[0, 2\pi)$?

Answer (c)

$\cos\theta = -\frac{1}{2}$ gives $\frac{2\pi}{3}, \frac{4\pi}{3}$.

$\cos\theta = 3$ has **no solution**: cosine is always between -1 and 1.

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

Note

FROM THE HANDOUT

Notice in the previous example, one of the equations did not provide a value of θ that satisfies the equation. This can happen sometimes!

Activity II

Activity II · from your notes

Solve each equation on the interval $[0, 2\pi)$.

(a) $2\cos^2\theta + \cos\theta = 0$

Answer (a)

GCF: $\cos\theta\,(2\cos\theta + 1) = 0$

- $\cos\theta = 0$ gives $\frac{\pi}{2}, \frac{3\pi}{2}$
- $\cos\theta = -\frac{1}{2}$ gives $\frac{2\pi}{3}, \frac{4\pi}{3}$

$$\theta = \frac{\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{3\pi}{2}$$

(b) $2\cos^2\theta - \cos\theta = 3$

> **Hint:** Get everything over to one side before you factor!

Answer (b)

Set it equal to zero: $2\cos^2\theta - \cos\theta - 3 = 0$, which factors as $(2\cos\theta - 3)(\cos\theta + 1) = 0$.

- $\cos\theta = \frac{3}{2}$: no solution (bigger than 1)
- $\cos\theta = -1$ gives π

$$\theta = \pi$$

Note

FROM THE HANDOUT

In this last activity, let's look at trigonometric equations that involve multiple angles.

Activity 12

Activity 12 · from your notes

Solve $\sin(2\theta) = \frac{1}{2}$. Find the general solution, then all solutions on $[0, 2\pi)$.

(a) Replace 2θ with α so it looks familiar. What values of α satisfy $\sin\alpha = \frac{1}{2}$ on $[0, 2\pi)$?

Answer (a)

$$\alpha = \frac{\pi}{6},\ \frac{5\pi}{6}$$

Careful: if θ runs from 0 to 2π, then $\alpha = 2\theta$ runs from 0 to 4π, which is two full turns. These two values cover only the first turn. The general solution in part (b) picks up the rest.

(b) What is the general solution of $\sin\alpha = \frac{1}{2}$?

Answer (b)

$$\alpha = \frac{\pi}{6} + 2\pi n \quad \text{and} \quad \alpha = \frac{5\pi}{6} + 2\pi n$$

(c) Since $\alpha = 2\theta$: $2\theta = \frac{\pi}{6} + 2\pi n$ and $2\theta = \frac{5\pi}{6} + 2\pi n$. Solve each for θ.

Hint: Solve for θ.

Answer (c)

Divide **every** term by 2, including the $2\pi n$.

$$\theta = \frac{\pi}{12} + \pi n \quad \text{and} \quad \theta = \frac{5\pi}{12} + \pi n$$

⚠ **Watch out:** A common slip is to divide only the first term and keep $+ 2\pi n$.

(d) Let $n = 0$. What values of θ result? Are they in $[0, 2\pi)$?

Answer (d)

$\frac{\pi}{12}$, $\frac{5\pi}{12}$: **both in the interval**

(e) Now let $n = 1$.

Answer (e)

Add $\pi = \frac{12\pi}{12}$.

$\frac{13\pi}{12}$, $\frac{17\pi}{12}$: **both in the interval**

(f) Suppose $n = 2$.

Answer (f)

$\frac{25\pi}{12}$, $\frac{29\pi}{12}$: **both are past** $2\pi = \frac{24\pi}{12}$, **so not in the interval**

(g) Now try $n = -1$.

Answer (g)

$\frac{-11\pi}{12}$, $\frac{-7\pi}{12}$: **negative, so not in the interval**

(h) Once you leave $[0, 2\pi)$ in a direction, stop. What are all the solutions?

Answer (h)

$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$

Four solutions: twice as many as $\sin\theta = \frac{1}{2}$ had.

Observation

FROM THE HANDOUT

Recall that $y = \sin(2\theta)$ is a horizontal compression by a factor of 2 of $y = \sin\theta$. On an interval of 2π we can graph **two** periods of $y = \sin(2\theta)$, as opposed to one cycle of $y = \sin\theta$. This suggests there may be twice as many solutions to $\sin(2\theta) = \frac{1}{2}$ compared to $\sin\theta = \frac{1}{2}$.

Practice Example 13

Practice Example 13 · from your notes

Find all solutions to $\cos(x) = \frac{1}{2}$ on the interval $[0, 2\pi)$.

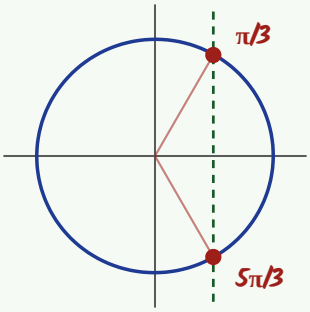
Try it, then check.

Answer

Cosine is the x-coordinate. It is positive in Quadrants I and IV, and the reference angle is $\frac{\pi}{3}$.

$x = \frac{\pi}{3}, \frac{5\pi}{3}$

; x = 1/2



Practice Example 14

Practice Example 14 · from your notes

Find the general solution to $\cos(2x) = \frac{\sqrt{2}}{2}$ (all solutions, not restricted to one period).

Try it, then check.

Answer

Let $\alpha = 2x$. $\cos\alpha = \frac{\sqrt{2}}{2}$ gives $\alpha = \frac{\pi}{4} + 2\pi n$ or $\alpha = \frac{7\pi}{4} + 2\pi n$.

Divide every term by 2:

$$x = \frac{\pi}{8} + \pi n \quad \text{and} \quad x = \frac{7\pi}{8} + \pi n, \quad n \text{ any integer}$$

Check: on $[0, 2\pi)$ these give $\frac{\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{15\pi}{8}$.

Practice Example 15

Practice Example 15 · from your notes

Find all solutions to $2\sin^2(x) - \sin(x) - 1 = 0$ on $[0, 2\pi)$.

Try it, then check.

Hint: Treat this as a quadratic in $\sin(x)$.

Answer

Let $u = \sin x$: $2u^2 - u - 1 = 0$ factors as $(2u + 1)(u - 1) = 0$. So $2\sin x + 1 = 0$ or $\sin x - 1 = 0$.

- $\sin x = -\frac{1}{2}$ gives $\frac{7\pi}{6}, \frac{11\pi}{6}$
- $\sin x = 1$ gives $\frac{\pi}{2}$

$$x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

Toolkit

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

Today's toolkit for trig equations

1. **Isolate** the trig function, then read angles off the unit circle.
2. **Square root?** Remember ±, which usually doubles the number of angles.
3. **Factor** (GCF or quadratic-style), then set each factor equal to zero. Never divide by a trig function.
4. **Reality check:** sin and cos can only equal values from −1 to 1.
5. **Multiple angle** like 2θ: substitute α, find the general solution, divide everything, then test values of n.
6. **General solution:** add $2\pi n$ to each solution in $[0, 2\pi)$.

Summary

FROM THE HANDOUT

Write 2–3 sentences summarizing the key concepts of today's lecture, in your own words.

Accessibility & help

These notes are built to work with screen readers, keyboard-only use, zoom and phone screens. Every answer stays hidden until you open it, and **Switch to standard font** at the top changes the handwriting to a plain font.

Other formats: If you need these notes another way (large print, a Word file, braille, or math written in MathML), message your instructor through Canvas Inbox, or contact Santa Ana College Disabled Students Programs & Services (DSPS): sac.edu/DSPS, DSPS@sac.edu, (714) 564-6295.

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